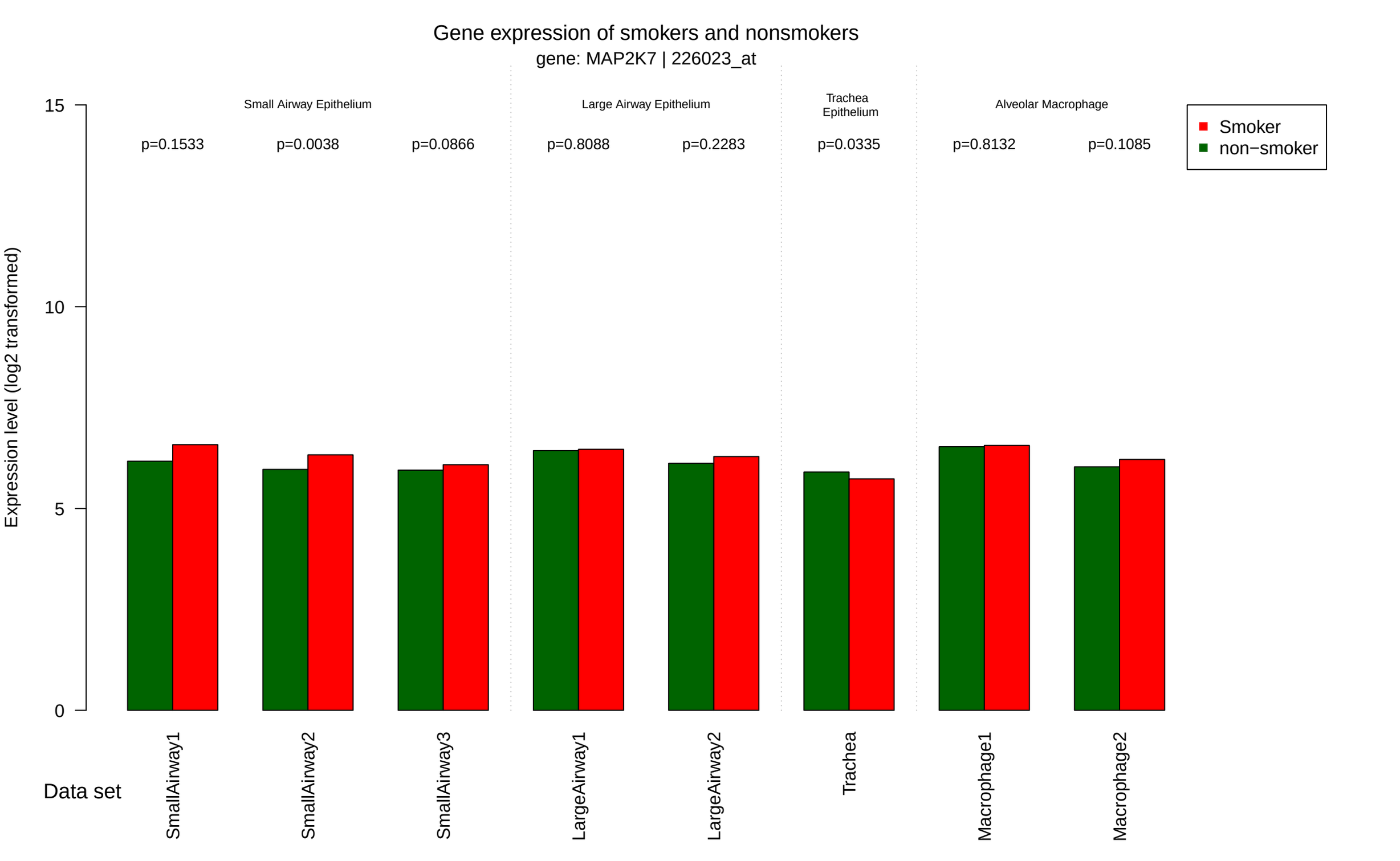
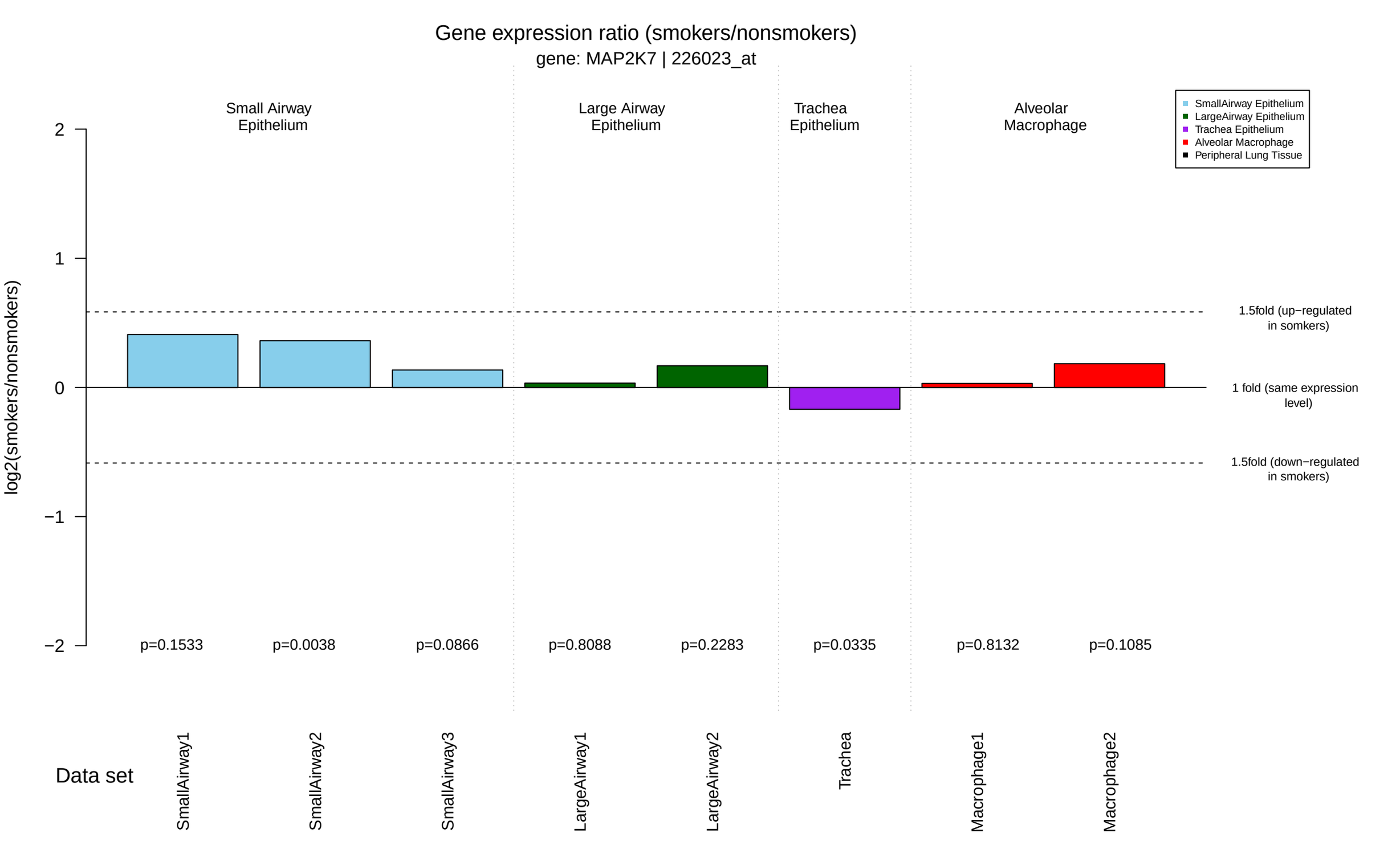
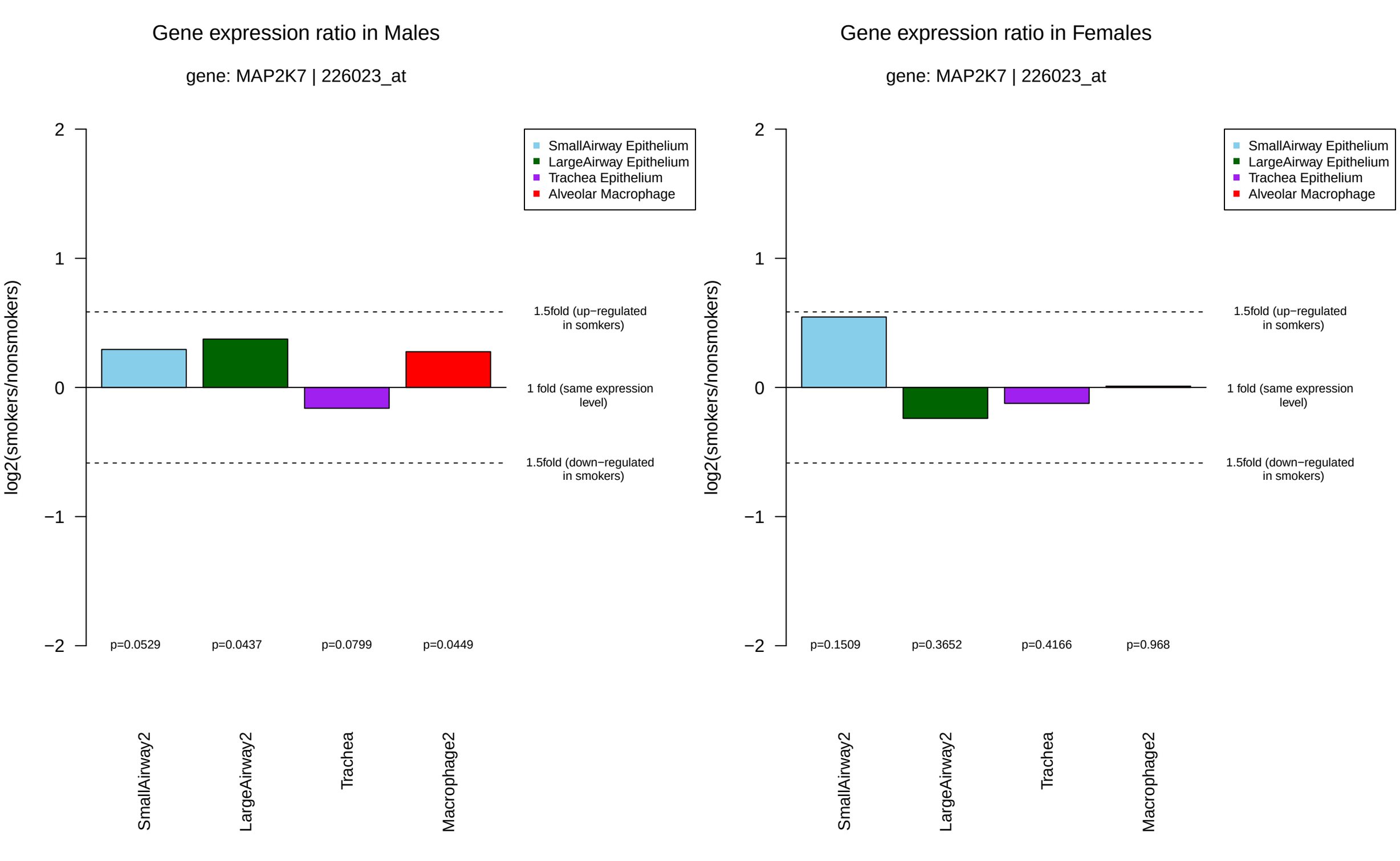
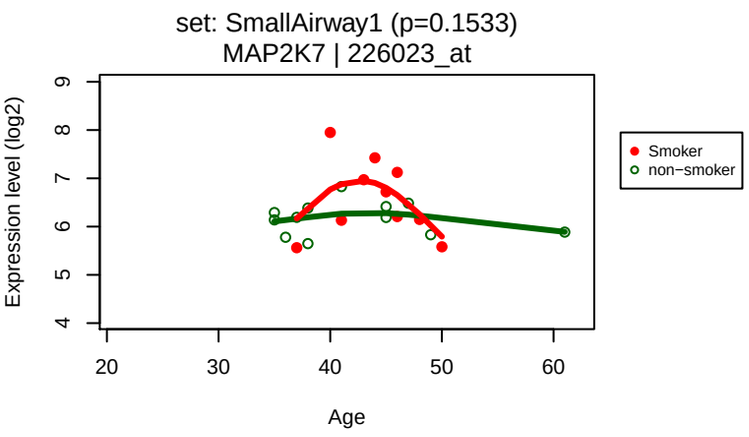
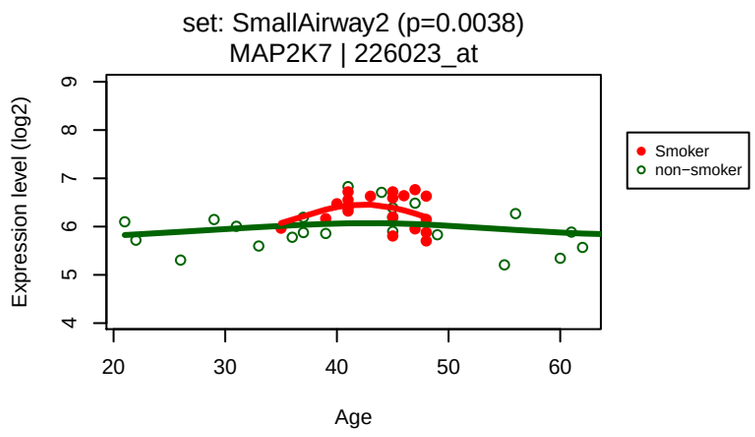
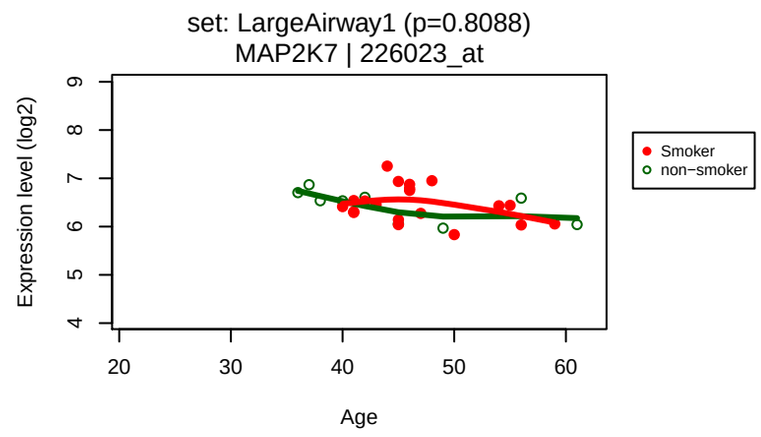

Gene expression of smokers and nonsmokers
gene: MAP2K7 | 226023_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1533
p=0.0038
p=0.0866
p=0.8088
p=0.2283
p=0.0335
p=0.8132
p=0.1085
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

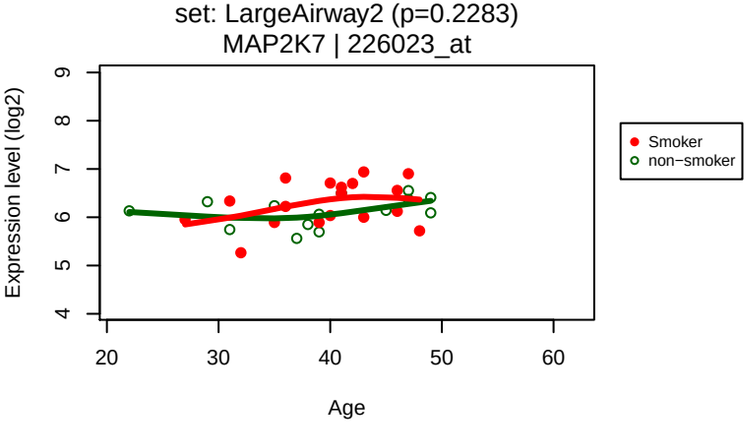

Gene expression ratio (smokers/nonsmokers)
gene: MAP2K7 | 226023_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1533
p=0.0038
p=0.0866
p=0.8088
p=0.2283
p=0.0335
p=0.8132
p=0.1085
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

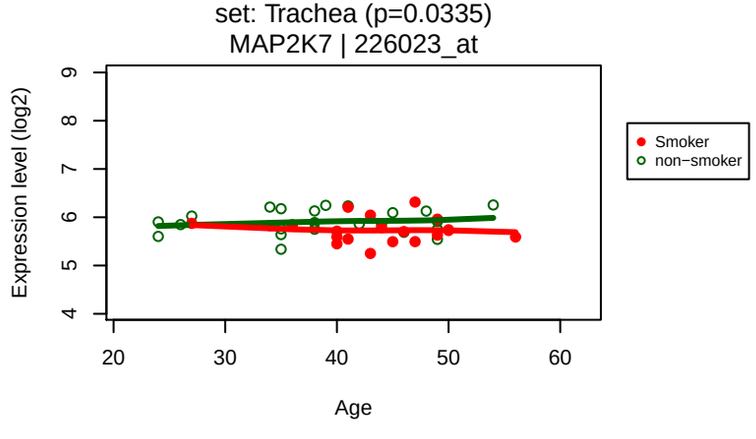

Gene expression ratio in Males
gene: MAP2K7 | 226023_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0529
p=0.0437
p=0.0799
p=0.0449
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAP2K7 | 226023_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1509
p=0.3652
p=0.4166
p=0.968
SmallAirway2
LargeAirway2
Trachea
Macrophage2

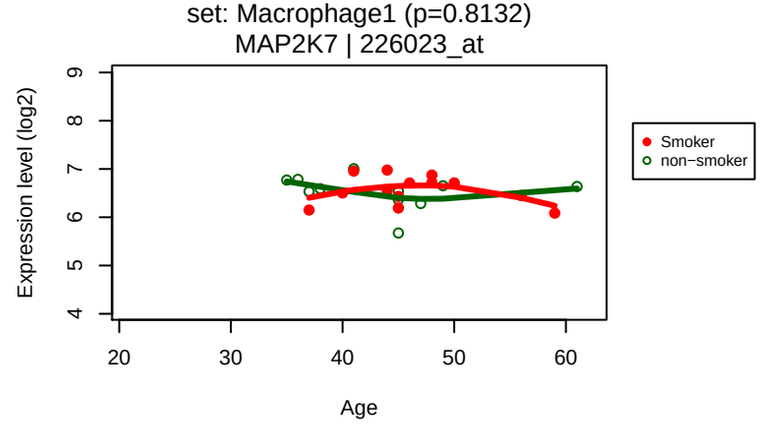

set: SmallAirway1 (p=0.1533)
MAP2K7 | 226023_at
Expression level (log2)
Age
Smoker
non−smoker

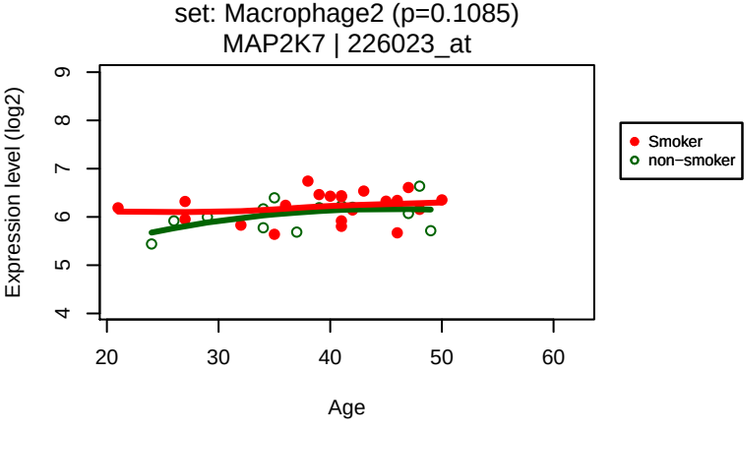

set: SmallAirway2 (p=0.0038)
MAP2K7 | 226023_at
Expression level (log2)
Age
Smoker
non−smoker

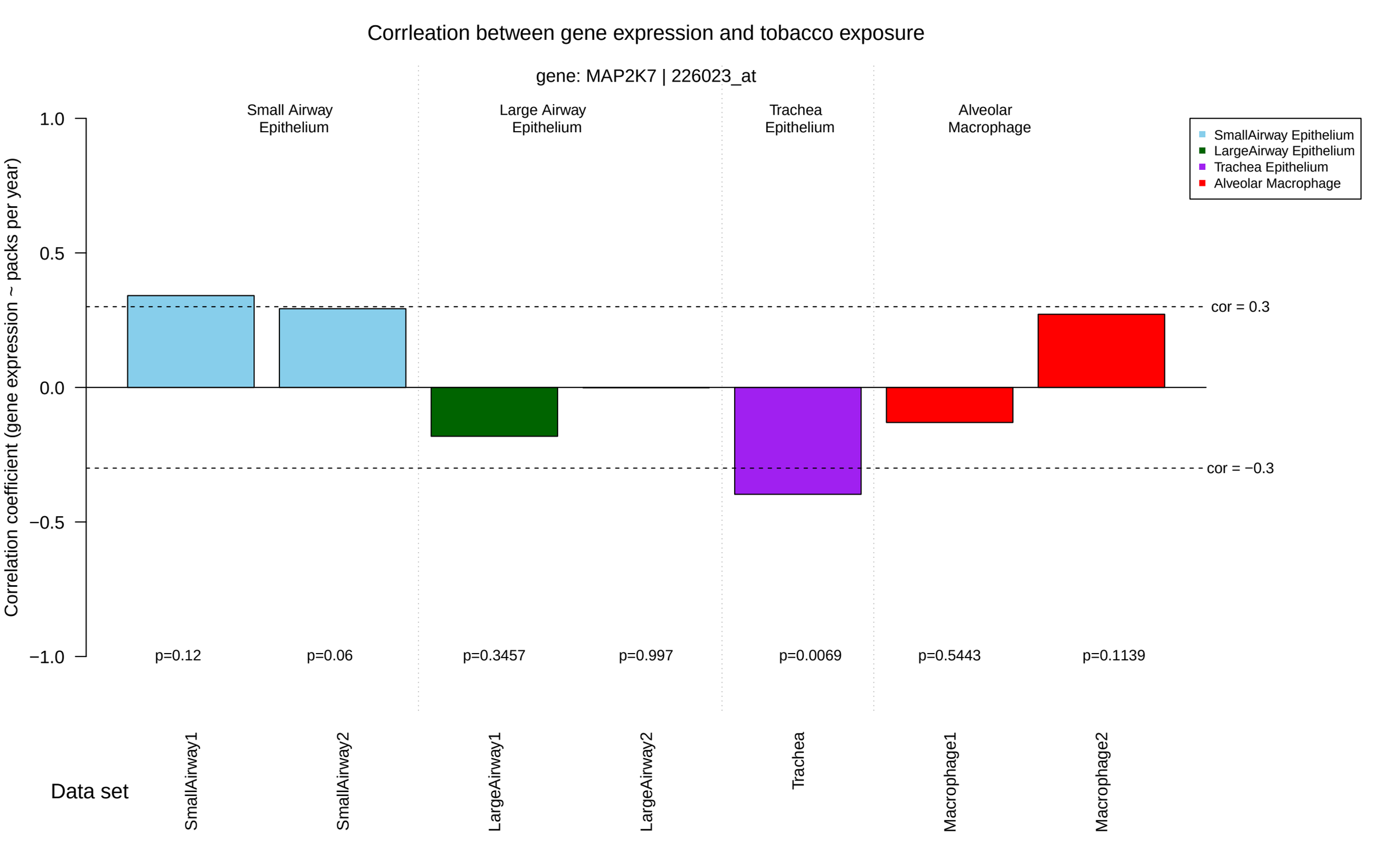

set: LargeAirway1 (p=0.8088)
MAP2K7 | 226023_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2283)
MAP2K7 | 226023_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0335)
MAP2K7 | 226023_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8132)
MAP2K7 | 226023_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1085)
MAP2K7 | 226023_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAP2K7 | 226023_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.12
p=0.06
p=0.3457
p=0.997
p=0.0069
p=0.5443
p=0.1139
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2